

PVsyst - Simulation report

Grid-Connected System

Project: caso3-denovo-manuas

Variant: New simulation variant

No 3D scene defined, no shadings

System power: 5.40 kWp

Manaus - Brazil

**PVsyst V8.1.5**

VC0, Simulation date:
13/09/26 12:50
with V8.1.5

Project summary**Geographical Site**

Manaus
Brazil

Situation

Latitude -3.1316 °(S)
Longitude -59.9825 °(W)
Altitude 43 m
Time zone UTC-4

Project settings

Albedo 0.20

Weather data

Manaus
NASA POWER satellite data 2001-2020 - Synthetic

System summary**Grid-Connected System**

No 3D scene defined, no shadings

Orientation #1
Horizontal plane

Near Shadings
no Shadings

User's needs
Unlimited load (grid)

System information**PV Array**

Nb. of modules 9 units
Pnom total 5.40 kWp

Inverters

Nb. of units 1 unit
Total power 5 kWac
Pnom ratio 1.08

Results summary

Simulation step Hourly

Produced Energy 7834.6 kWh/year Specific production 1451 kWh/kWp/year Perf. Ratio PR 85.09 %

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General parameters

Grid-Connected System

No 3D scene defined, no shadings

Simulation step

Hourly

Orientation #1

Horizontal plane

Models used

Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon

Free Horizon

Near Shadings

no Shadings

User's needs

Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer Generic
Model JKM-600N-72HL4-V
(Original PVsyst database)
Unit Nom. Power 600 Wp
Number of PV modules 9 units
Nominal (STC) 5.40 kWp
Modules 1 unit x 9 In series

At operating cond. (50°C)

Pmpp 5.01 kWp
U mpp 367 V
I mpp 14 A

Total PV power

Nominal (STC) 5 kWp
Total 9 modules
Module area 23.2 m²
Cell area 21.5 m²

Inverter

Manufacturer Generic
Model MIN 5000TL-X
(Original PVsyst database)
Unit Nom. Power 5.00 kWac
Number of inverters 1 unit
Total power 5.0 kWac
Operating voltage 80-550 V
Pnom ratio (DC:AC) 1.08
Power sharing within this inverter

Total inverter power

Total power 5 kWac
Number of inverters 1 unit
Pnom ratio 1.08

Array losses

Thermal Loss factor

Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses

Global array res. 440 mΩ
Loss Fraction 1.50 % at STC

Module Quality Loss

Loss Fraction -0.75 %

Module mismatch losses

Loss Fraction 2.00 % at MPP

Strings Mismatch loss

Loss Fraction 0.15 %

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000



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Main results

Simulation step

Hourly

System Production

Produced Energy

7834.6 kWh/year

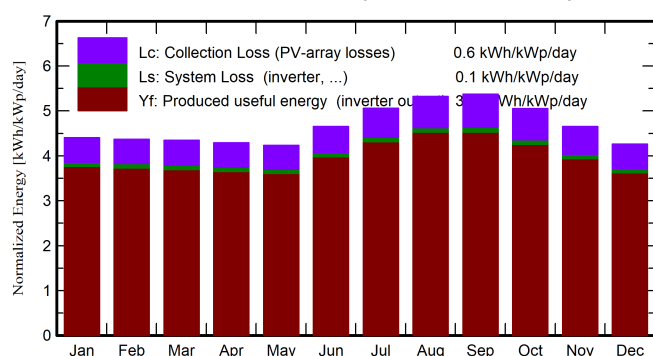
Specific production

1451 kWh/kWp/year

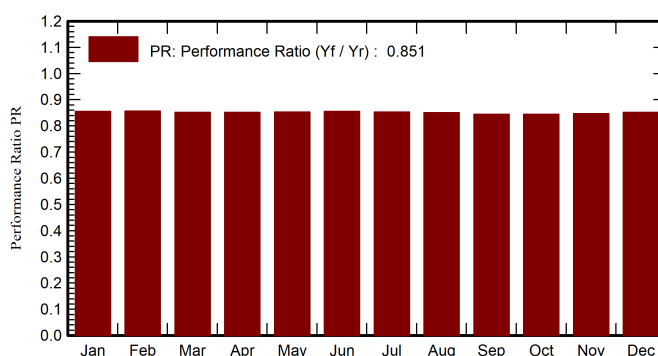
Perf. Ratio PR

85.09 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	DiffHor	T_Amb	GlobInc	GlobEff	EArray	E_Grid	PR
	kWh/m²	kWh/m²	°C	kWh/m²	kWh/m²	kWh	kWh	ratio
January	136.7	73.91	26.47	136.6	133.0	647.0	631.5	0.856
February	122.4	67.40	26.21	122.3	118.9	579.3	565.3	0.856
March	134.9	73.15	26.17	134.9	131.4	635.9	619.8	0.851
April	128.8	70.06	26.13	128.8	125.4	607.5	592.1	0.852
May	131.3	67.81	26.18	131.3	127.6	620.2	604.6	0.853
June	139.8	57.87	26.00	139.7	135.9	660.8	645.1	0.855
July	156.9	59.93	26.27	156.9	152.7	739.5	722.6	0.853
August	165.1	66.26	27.55	165.1	161.0	775.9	758.6	0.851
September	161.2	70.40	28.95	161.2	157.4	751.5	734.5	0.844
October	156.7	79.41	29.16	156.6	152.8	730.8	714.1	0.845
November	139.8	75.40	28.41	139.7	136.0	654.6	639.1	0.847
December	132.2	72.36	27.09	132.1	128.1	622.3	607.3	0.851
Year	1705.7	833.96	27.05	1705.1	1660.1	8025.4	7834.6	0.851

Legends

GlobHor Global horizontal irradiation

DiffHor Horizontal diffuse irradiation

T_Amb Ambient Temperature

GlobInc Global incident in coll. plane

GlobEff Effective Global, corr. for IAM and shadings

EArray Effective energy at the output of the array

E_Grid Energy injected into grid

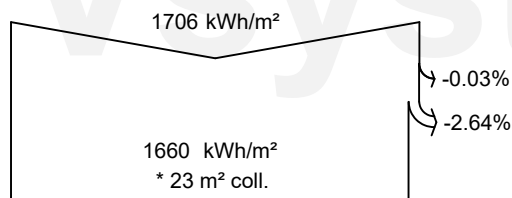
PR Performance Ratio



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Loss diagram



Global horizontal irradiation

Global incident in coll. plane

IAM factor on global

Effective irradiation on collectors

PV conversion

Array nominal energy (at STC effic.)

PV loss due to irradiance level

PV loss due to temperature

Module quality loss

Mismatch loss, modules and strings

Ohmic wiring loss

Array virtual energy at MPP

Inverter Loss during operation (efficiency)

Inverter Loss over nominal inv. power

Inverter Loss due to max. input current

Inverter Loss over nominal inv. voltage

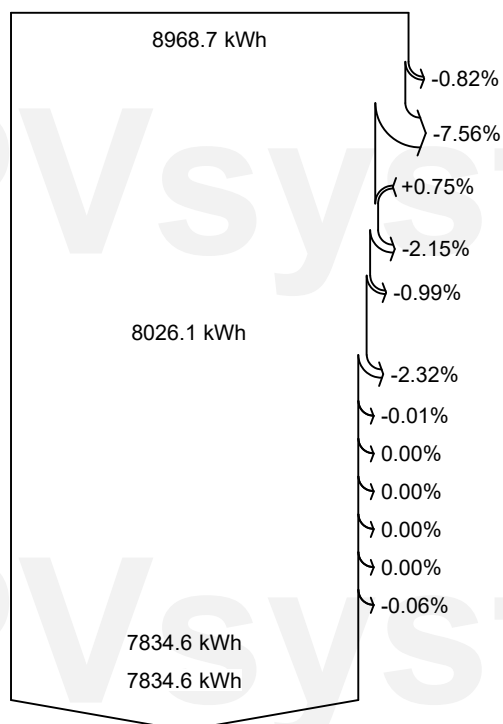
Inverter Loss due to power threshold

Inverter Loss due to voltage threshold

Night consumption

Available Energy at Inverter Output

Energy injected into grid



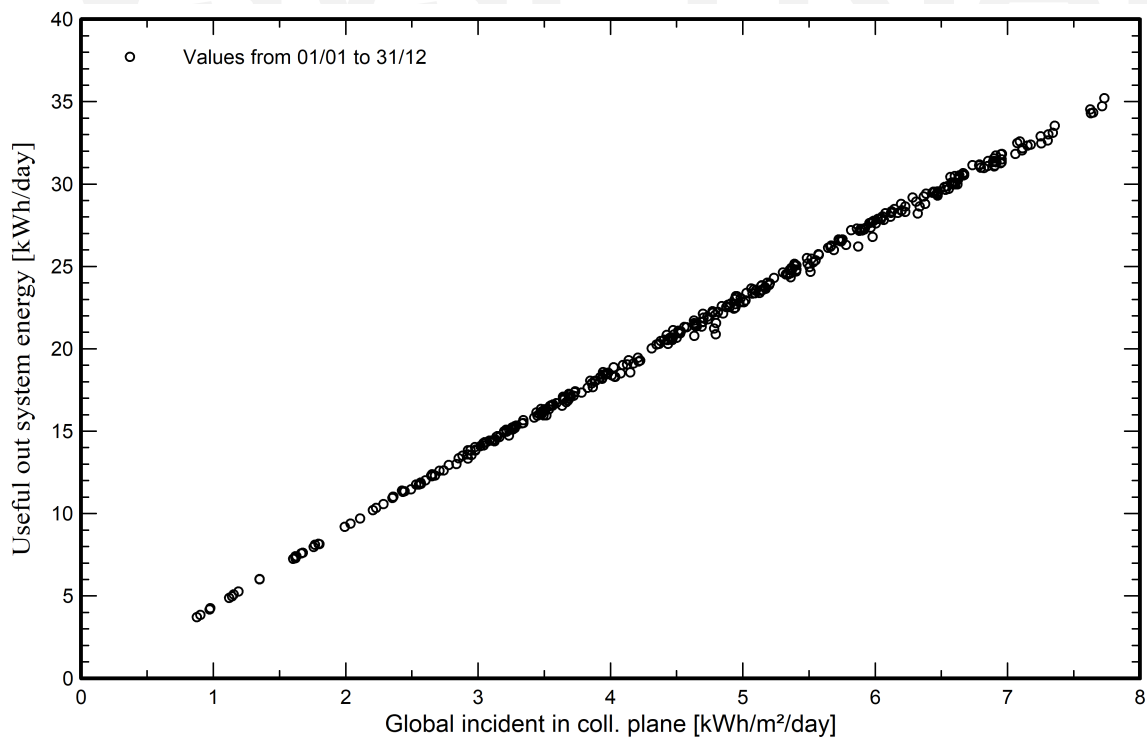


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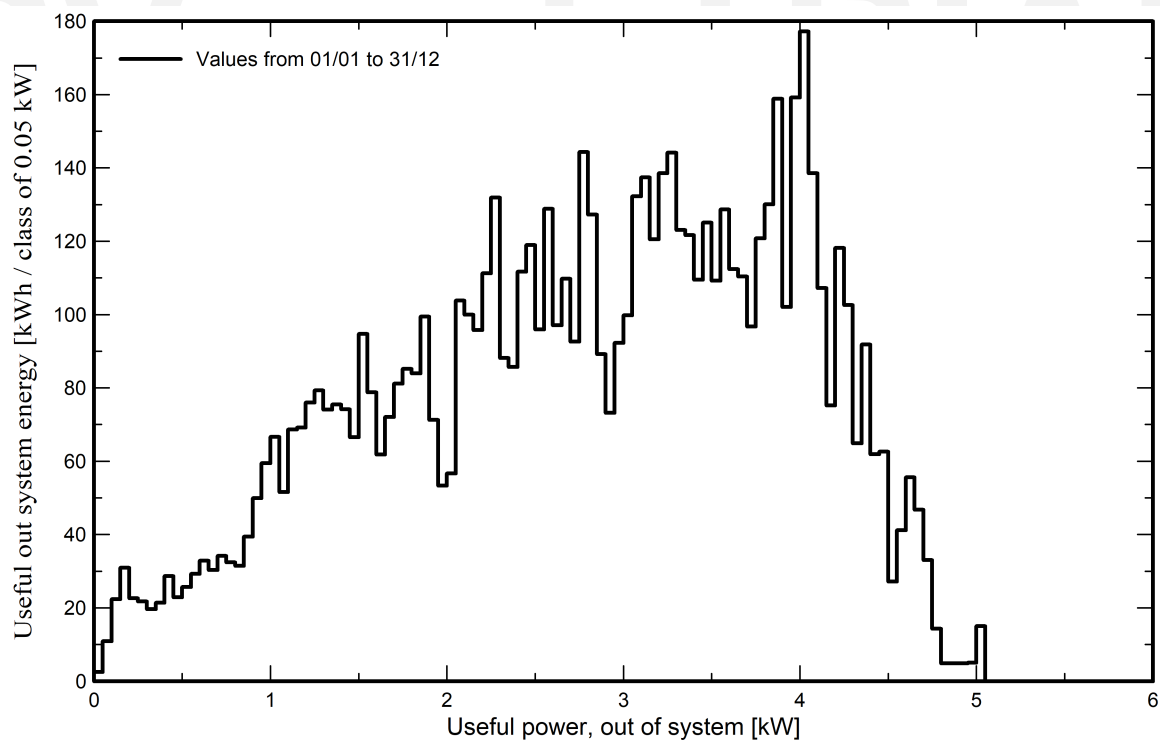
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Predef. graphs

Daily Input/Output diagram



System Output Power Distribution

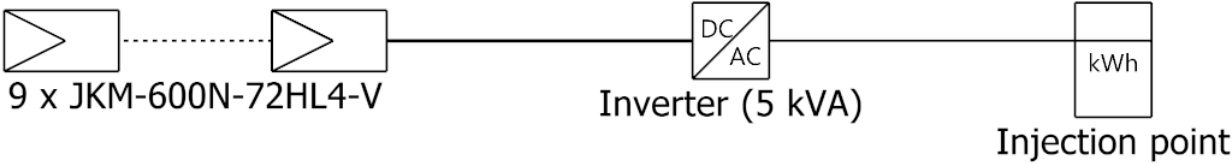




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Single-line diagram



PV module	JKM-600N-72HL4-V
Inverter	MIN 5000TL-X
String	9 x JKM-600N-72HL4-V

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